The Master Quadratic: One Equation, Two Crown Jewels

From the Diagonal of the GTE Polynomial to the Higgs Boson and Rule 110

UGP Physics Tutorial Series

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Contents

UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet*

See also: *The Levels of the Theory* · *How MDL Selects the 19-Bit Polynomial*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **$\mathbf{A}_{\text{Lean}}$ (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Diagonal: Setting $L = C = R = x$

Key idea

The polynomial p normally takes three *different* inputs. What if all three are the same? Setting $L = C = R = x$ reveals a hidden quadratic lurking inside the 3-body rule — the "master quadratic."

1.1 What "the diagonal" means

A cellular automaton is a row of cells, each with its own value. The polynomial $p(L, C, R)$ is built for the general case where Left, Centre, and Right all differ. But we can ask a special question: what happens on a perfectly *uniform* configuration — one where every cell holds the same value x?

Mathematically, this means setting $L = C = R = x$ and evaluating $p(x, x, x)$. This is called **evaluating on the diagonal** of the polynomial.

1.2 Computing $p(x, x, x)$ step by step

Substitute $L = C = R = x$ into $p(L, C, R) = C + R - CR - LCR$:

$$\begin{aligned} p(x, x, x) &= x + x - x \cdot x - x \cdot x \cdot x \\ &= 2x - x^2 - x^3. \end{aligned}$$

Nothing exotic: just algebra. The result is a cubic in x.

1.3 The self-consistency question

A natural follow-up: for which values of x does a uniform-x configuration *stay* uniform? Formally: when does $p(x, x, x) = x$? Such a value is called a **diagonal fixed point**.

Rearranging $p(x,x,x) = x$:

$$p(x,x,x) - x \;=\; 2x - x^2 - x^3 - x \;=\; x - x^2 - x^3 \;=\; -x\big(x^2 + x - 1\big).$$

So $p(x,x,x) = x$ if and only if $-x(x^2 + x - 1) = 0$, i.e. either $x = 0$ or $x^2 + x - 1 = 0$.

The universal diagonal factorization (machine-certified)

Over any number system,

$$\boxed{p(x,x,x) - x \;=\; -\,x\,(x^2 + x - 1).}$$

This identity is a universal algebraic fact, machine-certified in Lean 4 with zero sorry as `gte_diagonal_quadratic_factorization`. The factor $m(x) = x^2 + x - 1$ is the **master quadratic**.

The diagonal fixed-point equation has two families of solutions:

- $x = 0$: the *vacuum* — the all-zero configuration is always a fixed point.
- $m(x) = x^2 + x - 1 = 0$: solutions of the master quadratic (if any exist in the number system at hand).

How many solutions the master quadratic has — and in *which* number systems — governs two of the deepest results in the GTE framework.

2 The Discriminant: $\Delta = 5 = N_{\text{fam}}$

2.1 What is a discriminant?

For a quadratic $ax^2 + bx + c$, the **discriminant** is

$$\Delta = b^2 - 4ac.$$

It is the expression under the square root in the quadratic formula:

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

The discriminant determines whether the equation has solutions, and *where*:

- $\Delta > 0$: two distinct real roots.
- $\Delta = 0$: one repeated real root.
- $\Delta < 0$: no real roots (complex roots only).
- Δ is a quadratic non-residue mod q: no roots in the clock-arithmetic field $\mathrm{GF}(q)$.

The last case is central to Crown Jewel 2 in §4.

2.2 Computing the discriminant of $x^2 + x - 1$

For the master quadratic $m(x) = x^2 + x - 1$, we have $a = 1$, $b = 1$, $c = -1$:

$$\Delta = b^2 - 4ac = 1^2 - 4 \cdot 1 \cdot (-1) = 1 + 4 = \mathbf{5}.$$

2.3 The structural identity: $\Delta = N_{\text{fam}} = 5$

In the GTE framework, $N_{\text{fam}} = 5$ is the number of **fermion families** — the five particle types in a single Standard Model generation: electron, up quark, down quark, right-handed neutrino, left-handed neutrino. This number is derived independently from the $\mathbb{Z}_5$ orbit structure of the gauge sector (established in P01 [1]), not inserted by hand.

The identity $\Delta = N_{\text{fam}} = 5$ is structural: the family count is exactly the discriminant of the polynomial's diagonal self-consistency core. Every result that flows from the master quadratic — the Higgs VEV, the binary computational floor, vacuum uniqueness — inherits a link to the number of particle families.

Summary so far

$$p(x, x, x) - x = -x\,(x^2 + x - 1), \qquad \Delta = 5 = N_{\text{fam}}.$$

The master quadratic $x^2 + x - 1$ sits inside the diagonal of p. Its discriminant equals the fermion family count. Its behavior over $\mathbb{R}$ and over $\mathrm{GF}(7)$ are the two "crown jewels" of the framework.

3 Crown Jewel 1: The Golden Ratio and the Higgs VEV

Crown Jewel 1 — over $\mathbb{R}$

Over the real numbers, $x^2 + x - 1 = 0$ has a positive root $x^* = (\sqrt{5} - 1)/2 = 1/\varphi$, the reciprocal of the golden ratio. This is the SRRG fixed point — the self-consistent electroweak coupling — that seeds the Higgs vacuum expectation value $v = 246.16$ GeV.

3.1 Computing the real roots

Apply the quadratic formula to $x^2 + x - 1 = 0$ (so $a = 1$, $b = 1$, $c = -1$, $\Delta = 5$):

$$x = \frac{-1 \pm \sqrt{5}}{2}.$$

Root	Exact value	Approximation
$x^+ = \dfrac{-1+\sqrt{5}}{2}$	$= 1/\varphi$	≈ 0.6180
$x^- = \dfrac{-1-\sqrt{5}}{2}$	$= -\varphi$	≈ -1.6180

Here $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ is the **golden ratio**. The positive root $x^+ = (\sqrt{5} - 1)/2$ is exactly the reciprocal of the golden ratio, $1/\varphi = \varphi - 1$.

Verification: $(1/\varphi)^2 + (1/\varphi) - 1 = (\varphi - 1)^2 + (\varphi - 1) - 1 = \varphi^2 - 2\varphi + 1 + \varphi - 1 - 1 = \varphi^2 - \varphi - 1 = 0$ (since $\varphi^2 = \varphi + 1$ by the golden ratio property). ✓

3.2 The Self-Referential Renormalization Group

The **Self-Referential Renormalization Group** (SRRG) asks: is there an electroweak coupling g that is *self-consistent* with the universe's own dynamics? More precisely — in a universe whose fundamental rule is p — does there exist a coupling strength at which the dynamics, applied to itself, reproduces that same coupling?

This is a fixed-point question. Written in terms of the GTE polynomial, the SRRG fixed-point equation is:

$$p(g, g, g) = g.$$

From our computation, the non-trivial positive solution is:

$$g^* = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi} \approx 0.618.$$

The golden ratio is not an *assumption*; it is the *output* of the self-consistency equation of the master polynomial. Plain English: the golden ratio is the tuning fork that sets the electroweak scale — the unique self-consistent frequency at which the universe's own algebra resonates.

The Lean certification `gte_poly_srrg_bridge` (zero sorry) establishes that the SRRG fixed-point equation and the golden root of the master quadratic are one and the same object.

3.3 From the fixed point to the Higgs VEV

The **Higgs vacuum expectation value** v is the energy scale at which electroweak symmetry is broken, giving the $W^\pm$ and Z bosons their masses. Experimentally, $v \approx 246.22$ GeV (derived from the Fermi constant G_F).

In the Standard Model, v is a free parameter — it is fitted to data. In GTE, it is derived from the SRRG fixed point. The fixed-point coupling $g^* = 1/\varphi$, combined with the generation structure ($N_{\text{fam}} = 5$ from the $\mathbb{Z}_5$ orbit), gives:

$$v_{\text{PSC}} = 246.16 \text{ GeV} \qquad \text{(GTE prediction, machine-certified in Lean 4).}$$

The route from g^* to v runs through the SRRG renormalization-group cascade: the golden-ratio fixed point seeds the cascade, and the single dimensional anchor $m_\tau = 1776.86$ MeV combined with $N_{\text{fam}} = 5$ sets the energy scale; see P48 Chapter 7 for the explicit derivation. The agreement is $246.16/246.22 \approx 99.976\%$: a 0.024% discrepancy with zero free parameters.

Why the golden ratio?

The golden ratio satisfies $\varphi = 1 + 1/\varphi$ — it is self-similar by definition. In GTE, the electroweak coupling satisfies the analogous self-referential equation: $p(g, g, g) = g$. These are the same equation in disguise. The golden ratio is the algebraic fingerprint of self-reference in a universe governed by p.

4 Crown Jewel 2: Rootlessness over GF(7) and the Binary Floor

Crown Jewel 2 — over GF(7)

Over the integers mod 7, the equation $x^2 + x - 1 = 0$ has *no* solution. This rootlessness is the precise algebraic reason why the binary sector $\{0, 1\}$ is the *unique* minimal invariant sub-automaton of p — which is why Rule 110, restricted to binary inputs, is computationally universal.

4.1 Quadratic residues and non-residues

Recall from the Polynomial Cheat Sheet: in GF(7), all arithmetic is done modulo 7. Not every number has a square root in this finite world.

A **quadratic residue** mod 7 is a number that equals k^2 mod 7 for some $k \in \{0, 1, \ldots, 6\}$. A **quadratic non-residue** (QNR) is a number that is *not* the square of anything mod 7.

4.2 The squares of GF(7): a worked calculation

Compute k^2 mod 7 for every $k \in \{0, 1, 2, 3, 4, 5, 6\}$:

k	k^2 (ordinary)	k^2 mod 7	Remainder check
0	0	0	$0 = 7 \times 0 + 0$
1	1	1	$1 = 7 \times 0 + 1$
2	4	4	$4 = 7 \times 0 + 4$
3	9	2	$9 = 7 \times 1 + 2$
4	16	2	$16 = 7 \times 2 + 2$
5	25	4	$25 = 7 \times 3 + 4$
6	36	1	$36 = 7 \times 5 + 1$

The quadratic *residues* mod 7 — the values that appear in the "k^2 mod 7" column — are $\{0, 1, 2, 4\}$. The values $\{3, 5, 6\}$ never appear. In particular:

$$5 \text{ is a quadratic non-residue mod } 7.$$

This is a finite arithmetic fact, machine-certified as `five_is_qnr_mod7` (Lean 4, zero sorry, proved by `decide`).

4.3 Why 5 being a QNR means the quadratic has no root

The quadratic formula gives roots $x = (-1 \pm \sqrt{5})/2$. In GF(7), computing $\sqrt{5}$ requires finding s with $s^2 \equiv 5 \pmod 7$. But we just showed 5 is not among the squares mod 7 — so $\sqrt{5}$ does not exist in GF(7). The formula cannot produce a value, and the equation has no solution.

We can also verify directly by checking all seven candidates:

x	$x^2 + x - 1 \pmod 7$	Is it $\equiv 0$?
0	$0 + 0 - 1 = -1 \equiv 6$	No
1	$1 + 1 - 1 = 1$	No
2	$4 + 2 - 1 = 5$	No
3	$9 + 3 - 1 = 11 \equiv 4$	No
4	$16 + 4 - 1 = 19 \equiv 5$	No
5	$25 + 5 - 1 = 29 \equiv 1$	No
6	$36 + 6 - 1 = 41 \equiv 6$	No

None of the seven elements of GF(7) satisfies $x^2 + x - 1 = 0$. The master quadratic is **rootless** over GF(7).

4.4 Why rootlessness forces the binary floor

Return to the diagonal fixed-point equation: $p(x, x, x) = x$ has solutions $x = 0$ (vacuum) or $x^2 + x - 1 = 0$ (master quadratic). Over GF(7), the master quadratic has no solutions. Therefore the *only* constant diagonal fixed point of p over GF(7) is $x = 0$ — the vacuum.

Now ask: could some non-binary value $k \in \{2, 3, 4, 5, 6\}$ "anchor" a non-trivial invariant sub-automaton — a smaller set of states that the dynamics maps to itself, sitting between $\{0, 1\}$ and the full 7-state system? A necessary condition for k to do so is that $p(k, k, k) = k$ (the constant-k

configuration must be fixed). But the only constant fixed point is $k = 0$. No non-binary value qualifies.

Conclusion: the smallest non-trivial invariant subset under p in GF(7) is exactly $\{0, 1\}$ — the **binary floor**. The dynamics cannot produce an intermediate sub-automaton anywhere between the vacuum and the full 7-state system.

This is why Rule 110 is not merely *a* binary restriction of p but the *unique minimal computational layer*. The rootlessness of $x^2 + x - 1$ over GF(7) forces it.

Lean certifications

The QNR uniqueness theorem — $\{0, 1\}$ is the unique minimal invariant subset of p over GF(7) because $N_{\text{fam}} = 5$ is a quadratic non-residue mod 7 — is machine-certified in Lean 4 (zero sorry):

- `five_is_qnr_mod7` — the arithmetic fact: 5 is not a square mod 7
- `nfam_qnr_explains_binary_floor` — the structural implication

Module: `UgpLean/Universality/Z7InvariantSubsets.lean`.
The rootlessness is also 7-adically robust: $x^2 + x - 1$ has no root modulo 7^k for *any* power $k \geq 1$, certified by
`master_quadratic_no_root_mod_seven_pow`.

Complete invariant-subset taxonomy at $q = 7$

The p-invariant subsets of GF(7) — sets S such that $p(x, y, z) \in S$ whenever $x, y, z \in S$ — are *exactly*:

1. $\{0\}$ — the vacuum point.
2. $\{0, 1\}$ — the binary floor; the unique non-trivial proper invariant subset.
3. GF(7) itself — the full 7-state system.

No proper subset of size $3, 4, 5$, or 6 is p-invariant. The proof: any size-≥ 3 proper invariant set would require a non-binary constant fixed point $k \neq 0$ (i.e. $p(k, k, k) = k$), but $k = 0$ is the unique constant fixed point (the master quadratic has no root over GF(7)). The general statement — for any prime q where $x^2 + x - 1$ is rootless — is that the only p-invariant subsets are $\{0\}$, $\{0, 1\}$, and GF(q). This is a complete characterization; the analytic proof for all rootless primes is given in P49 [2].

5 The All-q Dichotomy Theorem

The rootlessness over GF(7) is not special to the prime 7. The splitting behaviour of $m(x) = x^2 + x - 1$ over GF(q) obeys a clean rule, the **all-q dichotomy theorem** (P49 [2], Theorem 4.4):

All-q dichotomy

For any prime $q \neq 2, 5$:

$$x^2 + x - 1 = 0 \text{ has roots in } \mathrm{GF}(q) \iff 5 \text{ is a QR mod } q \iff q \equiv \pm 1 \pmod 5.$$

Rootless (QNR) branch ($q \equiv \pm 2 \pmod 5$, e.g. $q = 7, 13, 37, \ldots$): $\{0, 1\}$ is the unique binary floor; Rule 110 universality is forced.

Split (QR) branch ($q \equiv \pm 1 \pmod 5$, e.g. $q = 11, 19, 29, \ldots$): the quadratic has two roots; additional invariant subsets emerge.

Ramified case ($q = 5$): discriminant vanishes, the quadratic is $(x+3)^2$ with double root $x = 2$. (This is the "displaced vacuum" pathology of the GF(5) world.)

The prime $q = 7$ satisfies $7 \equiv 2 \pmod 5$, placing it firmly on the rootless branch — the unique class of primes where $N_{\text{fam}} = 5$ obstructs any intermediate invariant sub-automaton.

Prime q	q mod 5	5 is...	Master quadratic over GF(q)
2	2	special	Irreducible
5	0	Ramified	Double root $(x + 3)^2$
7	2	**QNR**	**Rootless** ← our universe
11	1	QR	Two distinct roots
13	3	QNR	Rootless
19	4	QR	Two distinct roots
29	4	QR	Two distinct roots
37	2	QNR	Rootless

The prime 7 is not chosen because it gives a nice result; it is determined by independent constraints (the ridge minimality theorem of P01). The QNR condition is then verified as a *consequence*. The computational floor is a theorem, not a design choice.

6 One Quadratic, Two Scales

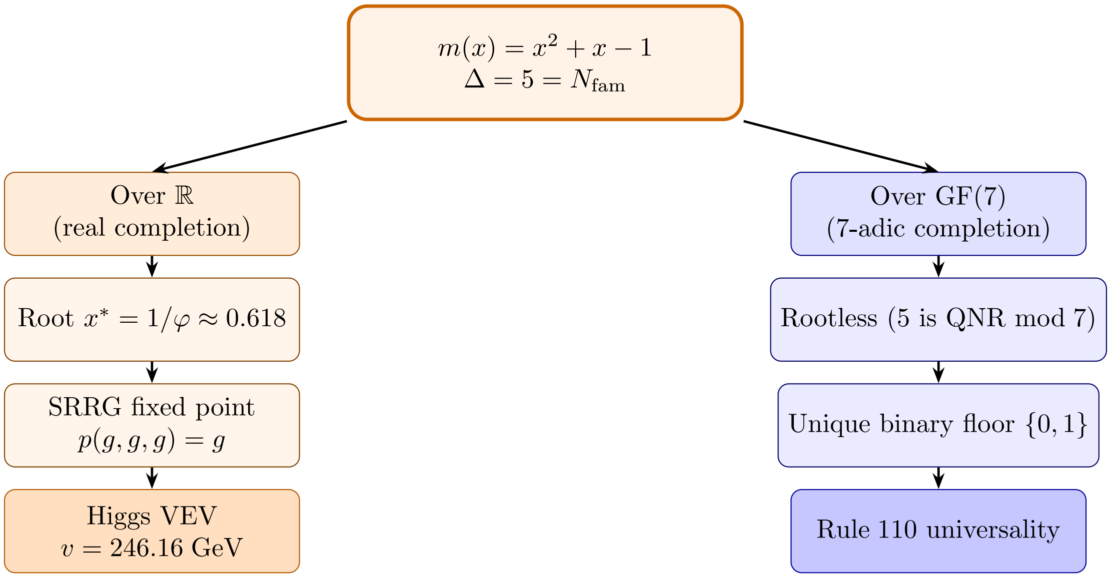
6.1 The unification

The same $\mathbb{Z}$-polynomial $m(x) = x^2 + x - 1$ — discriminant $5 = N_{\text{fam}}$ — controls two previously separate derivation chains via its two natural arithmetic completions:

- **Over** $\mathbb{R}$ (archimedean completion): the polynomial splits, yielding the golden root $1/\varphi$ that is the SRRG fixed point and seeds the Higgs VEV $v = 246.16$ GeV.
- **Over GF(7)** (7-adic completion): the polynomial is inert, with no roots — the mechanism that makes $\{0, 1\}$ the unique binary floor and forces Rule 110 computational universality.

These are not two separate facts that happen to share a quadratic. They are both arithmetic images of the single integer polynomial m under the two completion maps $\mathbb{Z}[x] \to \mathbb{R}[x]$ and $\mathbb{Z}[x] \to \mathrm{GF}(7)[x]$. This unification is the *Golden-Quadratic Duality* theorem (P49 §3.2, CatAD; core components CatAL).

6.2 Diagram



6.3 Comparison table

	Over $\mathbb{R}$	**Over** GF(7)
$\sqrt{5}$ exists?	Yes	No (5 is a QNR)
Roots of $x^2 + x - 1$?	Yes: $1/\varphi$ and $-\varphi$	None
Fixed-point eq. $p(x, x, x) = x$	golden root $1/\varphi$	vacuum only ($x = 0$)
Physical role	SRRG coupling	Unique binary floor
Physical output	$v = 246.16$ GeV	Rule 110 universality
Lean cert	`gte_poly_srrg_bridge`	`five_is_qnr_mod7`

7 Vacuum Uniqueness

A third result from the same quadratic

The master quadratic controls a third independent result: on a ring of *any* size n, the vacuum (all-zero configuration) is the *unique* temporally fixed configuration of the p-dynamics. There is only one way the GTE universe can "do nothing."

The GTE polynomial runs on a **ring**: a row of cells wrapped end-to-end so the last cell is adjacent to the first. On a ring of size n, we can ask: which configurations are **temporally fixed** — unchanged when every cell updates simultaneously according to p?

The vacuum (every cell holds value 0) is always a fixed point: $p(0, 0, 0) = 0$.

Are there other temporally fixed configurations? One might hope so — perhaps some periodic pattern on a sufficiently large ring is also left unchanged. The answer, from the master quadratic, is *no*.

Any non-vacuum temporally fixed configuration would require some cell to hold a value $k \neq 0$ in a context where $p(k, k, k) = k$ (the simplest version of the argument; the full proof covers all ring geometries and uses the Fibonacci-Möbius map associated to the dynamics). The master quadratic $x^2 + x - 1$ has no root over GF(7), so no such k can exist.

The theorem:

$$\mathrm{Fix}(T_n) = \{\text{vacuum}\} \quad \text{for every ring size } n.$$

Machine-certified as `vacuum_unique_temporal_fixed_point_ring` (Lean 4, zero sorry, module `UgpLean/Polynomial/DynamicalZeta.lean`).

This is the *dynamical* complement of the algebraic uniqueness. The spatial version says the vacuum is the unique pointwise fixed state of the GF(7) rule (`vacuum_unique_fixed_point_z7`); the temporal version says it is the unique temporally fixed configuration of the full ring dynamics at every ring size.

Plain-English summary

There is only one frozen configuration in the GTE universe: the vacuum. Any non-trivial pattern — even on a huge ring — will eventually evolve. The master quadratic is the algebraic reason: rootlessness over GF(7) blocks every alternative frozen configuration.

8 Why This Is Remarkable

The Higgs mechanism and computational universality look, on the surface, like phenomena from completely separate intellectual territories:

- The **Higgs mechanism** explains why the $W^{\pm}$ and Z bosons are massive, and hence why the weak nuclear force is so short-ranged. It is a statement in relativistic quantum field theory at energy scales around 250 GeV.
- **Computational universality** (Rule 110) is a theorem in theoretical computer science: a binary cellular automaton — a row of zeros and ones updated by a simple rule — can simulate any Turing machine. It lives in the realm of discrete mathematics and computability theory.

In virtually every framework before GTE, these two facts occupy entirely different branches of physics and mathematics. No connection between them was expected, let alone known.

In GTE, both follow from one integer quadratic, x^2+x-1, whose discriminant is the number of fermion families:

- The **real root** $1/\varphi$ is the SRRG fixed point. Combined with the generation structure, it gives $v = 246.16$ GeV.
- The **rootlessness** over GF(7) (because 5 is a QNR mod 7) forces $\{0, 1\}$ as the unique minimal invariant floor, making Rule 110 the unique binary sub-layer and inheriting its universality.

The connection is not a post-hoc observation — it is a machine-certified structural theorem, the Golden-Quadratic Duality (`golden_floor_duality_bundle`, Lean 4, zero sorry).

The quadratic itself is not inserted by hand. It emerges from the diagonal of the master polynomial p, which is in turn selected by minimum description length from among 10^{290} candidate CA rules (established in P40/P48 [3]). One arithmetic object, two completions, two scales — electroweak and computational — unified at the algebraic level.

A further algebraic layer comes from the prime behaviour of $q = 7$ with respect to 2. The prime 2 is *inert* in $\mathbb{Z}[\varphi]$ (it does not split), so the degree-2 extension GF(4) sits as the additive group $V_4 \cong \mathrm{GF}(4)^+$ inside $A_4 = V_4 \rtimes \mathbb{Z}_3$, the alternating group on four letters. The prime 7 is *split* in $\mathbb{Z}[\varphi]$, generating the Frobenius group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ as the GTE orbit-stabilizer group. The leading-order neutrino mixing (tribimaximal) arises from A_4 as the discrete flavor symmetry at the inert prime, while F_{21} controls the generation structure at the split prime. Both algebraic structures trace to the single quadratic $x^2 + x - 1$ and the two ways its splitting field interacts with the small primes 2 and 7.

The splitting-field structure carries a third consequence, for quantum mechanics. Since $[\mathrm{GF}(49) : \mathrm{GF}(7)] = 2$ and the Frobenius automorphism $x \mapsto x^7$ swaps the two roots of $m(x)$ by complex conjugation, the continuum amplitude field is forced to $\mathbb{C}$ — the unique degree-2 extension of $\mathbb{R}$ — providing a structural derivation of the complex number field of quantum mechanics from the master quadratic (P48 [3] § Why Complex QM, CatAD).

The master quadratic at a glance

Object	Description
Origin	Diagonal of p: $p(x, x, x) - x = -x(x^2 + x - 1)$
Discriminant	$\Delta = 5 = N_{\text{fam}}$ (Standard Model fermion family count)
Over $\mathbb{R}$	Positive root $1/\varphi$ (golden ratio reciprocal)
Over GF(7)	Rootless (5 is a QNR mod 7)
Crown Jewel 1	SRRG fixed point $\to$ Higgs VEV $v = 246.16$ GeV
Crown Jewel 2	Unique binary floor $\{0, 1\}$ $\to$ Rule 110 universality
Bonus	$\mathrm{Fix}(T_n) = \{\text{vacuum}\}$ for all ring sizes n

Further Reading

Primary Sources

The results in this tutorial are derived and machine-certified in the following papers, all open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (synthesis monograph; §4 is the master-quadratic chapter with all Lean certifications):
doi.org/10.5281/zenodo.20560550
- **P49 — MDL Selects the Wolfram Rule** (Golden-Quadratic Duality theorem, QNR floor uniqueness, all-q dichotomy; §3):
doi.org/10.5281/zenodo.20572656
- **P27 — The Self-Referential Renormalization Group** (full derivation of $v = 246.16$ GeV from the SRRG golden-ratio fixed point):
doi.org/10.5281/zenodo.20170859

Lean 4 machine proofs: github.com/novaspivack/ugp-lean
modules: `GoldenQuadratic.lean`, `Z7InvariantSubsets.lean`, `SRRGCABridge.lean`, `DynamicalZeta.lean`.

References

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[2] Nova Spivack. MDL selects the Wolfram rule: $\mathbb{Z}_7$ dynamics, algebraic structure, and Standard Model encoding of the GTE polynomial. Preprint, 2026. URL https://doi.org/10.5281/zenodo.20572656. https://doi.org/10.5281/zenodo.20572656. UGP Physics series, P49.

[3] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL https://doi.org/10.5281/zenodo.20560550. https://doi.org/10.5281/zenodo.20560550. Preprint.